MILITARY SPECIFICATION

FUZE, POINT INITIATING, BASE DETONATING, M509A1E1, LOADING ASSEMBLING AND PACKING

This Specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type fuze designated as Fuze, Point Initiating, Base Detonating, M509A1E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or requests for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, MEthods of
MIL-A-48078 - Ammunition Standard Quality Assurance Provisions General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-650 - Explosive: Sampling, Inspection and Testing

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARMY

8799735 - Fuze, PI, BD, M509A1E1 Assembly
8837807 - Box, for Fuze, Point Initiating, Base Detonating, M509A1, or Fuze Point Initiating, Base Detonating, M530
9248976 - Carton Packing, For Fuze, PI, BD, M530

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts and assemblies.-The parts and assemblies shall comply with all requirements specified on Drawing (dwg.) 8799735 and with all requirements specified in applicable specifications.

3.3 Non-arming (prior to loading.-The metal parts assembly, with detonator, shall not arm when subjected to a minimum (min.) centrifugal acceleration (see 6.5) of 2500 gravities (g's), when tested as specified in 4.4.4.

3.4 Functioning.-There shall be no evidence of a premature burst and the fuze shall function the projectile high order as evidenced by penetration of the armor plate target by jet action, when tested as specified in 4.4.5 (see 6.7).

3.5 Workmanship.-All parts shall be fabricated and finished in a thorough workmanlike manner. The parts shall be free of burrs, chips, cracks, sharp edges, unblended radii, surface defects and dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are part of the details specified.

4.2 Classification of inspection.-The following types of inspections shall be conducted on this item.

a. Production quality conformance inspection.

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4.3 Quality conformance inspection

4.3.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Explosive material for lead charges from not more than two consecutive lots with the same lot interfix number from one manufacturer.

c. Explosive material for booster pellets from not more than two consecutive lots with the same lot interfix number from one manufacturer.

4.3.2 Examination.-One hundred percent examination shall be performed for all Critical defects and Major defects when so specified. Examination for Major and Minor defects shall be performed on a defect basis in accordance with the Classification of Defects contained herein using the sampling plans of MIL-STD-105, Level II. All non-conforming material shall be rejected, see MIL-A-48078.

NOTE: Critical Defect Reporting - A report of all Critical Defects found, number of parts or assemblies inspected for Critical defects, and corrective action taken, if any, shall be compiled monthly for the duration of the contract and copies forwarded to the Contracting Officer and Commander, ARRADCOM Hqts, Dover, NJ 07801, ATTN: DRDAR-LCR-E and DRDAR-QA-A-F.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Booster Lead Cup Assembly				8799767
					NEXT HIGHER ASSEMBLY
					9323859
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Cup split or cracked.		100%	3.2	Visual
Major					
101	Depth to charge, max.		0.40%	3.2	Gage
102	Charge above flush.				
Minor					
201	Evidence of poor workmanship.		0.65%	3.5	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 71 which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.2	Booster Pellet		9323860
			NEXT HIGHER ASSEMBLY 8799735

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Diameter, min.		0.40%	3.2	Gage
102.	Thickness.		0.40%	3.2	Gage
103.	Cracked or split.		0.40%	3.2	Visual
Minor	None Defined.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.3	Shield Loading Assembly		9323859
			NEXT HIGHER ASSEMBLY 8799735

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Booster lead cup protrudes beyond surface of release mechanism cavity.		0.40%	3.2	Gage
102.	Booster lead cup assembly missing.		0.40%	3.2	Visual
103.	Booster lead cup not crimped full 360° (when applicable).		0.40%	3.2	Visual
104.	Foreign matter in release mechanism cavity or booster pellet cavity.		0.40%	3.2	Visual
105.	Depth to charge from bottom surface of booster cavity.		0.40%	3.2	Gage
Minor					
201.	Evidence of poor workmanship.		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.4	Fuze Assembly Prior to Assembly of Shield				8799735
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Fuze in armed position.		100%	3.2	Visual
2.	Rotor spring not fully engaged in rotor slot.		100%	3.2	Visual
3.	Insulator resistor assembly missing.		100%	3.2	Visual
4.	Release mechanism loose.		100%	3.2	Manual
Major	None Defined.				
Minor					
201.	Evidence os poor workmanship.		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.5	Fuze Assembly, Prior to Assembly of Booster Cover		8799735
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Booster pellet split, cracked or edges chipped.		0.40%	3.2	Visual
Minor					
201.	Evidence of poor workmanship.		0.65%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.6	Fuze Assembly				8799735
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Position of spring pin.		0.40%	3.2	Gage
102.	Total length, max.		0.40%	3.2	Gage
103.	Outside diameter below flange, max.		0.40%	3.2	Gage
104.	Diameter of flange, max.		0.40%	3.2	Gage
105.	Shield loose.		0.40%	3.2	Manual
106.	Booster cover loose.		0.40%	3.2	Manual
107.	Protective finish damaged exposing base metal.		0.40%	3.2	Visual
Minor					
201.	Marking missing, incorrect or unidentifiable.		0.65%	3.2	Visual
202.	Evidence of poor workmanship.		0.65%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.7	Carton Packing, Prior to Sealing		9248976
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Fuze missing.		0.40%	3.2	Visual
102.	Separator damaged to the extent that fuzes touch each other.		0.40%	3.2	Visual
103.	Any packing material missing.		0.40%	3.2	Visual
Minor					
201.	Evidence of poor workmanship.		0.65%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.8	Carton, Packing, Sealed		9248976
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Carton torn, cut or punctured.		0.40%	3.2	Visual
102.	Sealing strip torn, badly wrinkled, incomplete or fails to adequately seal box.		0.40%	3.2	Visual
Minor					
201.	Marking misleading or unidentifiable.		0.65%	3.2	Visual
202.	Evidence of poor workmanship.		0.65%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.9	Sealed barrier bag		9248976
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Seal improper or incomplete.		0.40%	3.2/3.5	Visual
102.	Bag punctured, torn or cut.		0.40%	3.2/3.5	Visual
103.	Heat seal test failure.			3.2/3.5	Test - 4.3.3.7
Minor					
201.	Evidence of poor workmanship.		0.65%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.10	Box Packing, Prior to Sealing				8837807
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Carton missing.		0.40%	3.2	Visual
Minor					
201.	Evidence of poor workmanship.		0.65%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63270 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.10	Packing Box, Sealed		8837807
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Box broken, board loose, or protruding nail.		0.40%	3.2	Visual
102.	Hardware or strapping missing broken or loose.		0.40%	3.2	Visual
Minor					
201.	Car seal missing, unsealed or improperly positioned.		0.65%	3.2	Visual
202.	Marking misleading or unidentifiable.		0.65%	3.2	Visual
203.	Hardware or strapping improperly engaged or mislocated.		0.65%	3.2	Visual-Manual
204.	Contents loose.		0.65%	3.2	Visual-Manual
205.	Evidence of poor workmanship.		0.65%	3.5	Visual

NOTES:

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4.3.3 Testing

4.3.3.1 Non-arming - (see 3.3) Critical defect.-This test shall be performed 100 percent. The test shall be performed as specified in 4.4.4, using test equipment specified in 4.3.4. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot and the finding immediately reported.

4.3.3.2 Determination of specific gravity of booster pellet - (see dwg. 9323860) Major defect.-Ten (10) booster pellets shall be selected from each lot for this test. The test shall be performed as specified in 4.4.1. If one or more booster pellets fail to comply with the specified drawing requirement the entire lot of booster pellets represented by the sample shall be rejected.

4.3.3.3 Determination of binder-lubricant content of booster lead charge - (see dwg. 8799767) Major defect.-The percentage of binder-lubricant used in the booster lead cup charge shall be determined for each lot at the time of loading. Approximately twenty (20) grams shall be selected from each lot for this test. If the sample does not comply with the specified drawing requirement the entire lot represented by the sample shall be rejected. The test shall be performed as specified in 4.4.3.

4.3.3.4 Determination of specific gravity of booster lead charge.-For the following tests, the batch as defined in 4.4.2 shall retain the identity assigned therein.

4.3.3.4.1 Preformed pellets (if applicable) - (see dwg. 8799767) Major defect.-A sample of twenty (20) units shall be selected from each batch for this determination which shall be performed as specified in 4.4.2.1. If one or more samples fail to conform to the specified requirement, the entire batch represented by the sample shall be rejected.

4.3.3.4.2 Reconsolidated pellet in lead cup - (see dwg. 8799767) Major defect.-A sample of twenty (20) loaded lead cups shall be selected from each batch for this determination which shall be performed as specified in 4.4.2.2. If one or more samples fail to conform to the specified requirement the entire batch represented by the sample shall be rejected.

4.3.3.5 Functioning.-These tests shall be performed at a Government Proving Ground.

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4.3.3.5.1 Plate performance (see 3.4) - Premature burst - Critical defect, plate penetration failure - Major defect.-When the fuzes are to be used in Cartridges 105MM, HEAT-T, M456A1E1, 90MM, HEAT-T, M431, they shall be tested in Cartridges 105MM, HEAT-T, M456A1E1. When the fuzes are to be used in Cartridge, 106MM, HEAT, M344A1, they shall be tested in that cartridge. Testing shall be performed as specified in 4.4.5.

4.3.3.5.2 Functioning in cartridge, 105MM, HEAT-T, M456A1E1.-Twenty (20) fuzes shall be selected at random from each lot for this test. If four (4) or more samples fail to comply with the specified penetration requirement the lot shall be rejected. If two (2) or three (3) samples fail, a second sample of forty (40) shall be selected and tested. If in the first and second sample combined the total number of penetration failures is four (4) or more the lot shall be rejected. If a premature burst (in bore or prior to the normal arming distance of the fuze), occurs at any time, the lot shall be rejected and the finding immediately reported.

4.3.3.5.3 Functioning in Cartridge, 106MM, HEAT, M344A1.-Twenty (20) fuzes shall be selected at random from each lot for this test. If four (4) or more samples fail to comply with the specified penetration requirement the lot shall be rejected. If two (2) or three (3) samples fail, a second sample of forty (40) shall be selected and tested. If in the first and second sample combined the total number of penetration failures is four (4) or more the lot shall be rejected. If a premature burst occurs at any time the lot shall be rejected and the finding immediately reported.

4.3.3.6 Heat seal of sealed bag (see dwg 9248976).-Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-P-116.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and test prescribed herein is described in the Paragraph/Inspection Method column in the tables starting with Paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment design in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.4 TEST METHODS AND PROCEDURES

4.4.1 Determination of specific gravity of booster pellet.-The specific gravity of the booster pellet shall be determined in accordance with Method 203.1 of Standard MIL-STD-650. The

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pellet shall be classed defective if the specific gravity fails to conform to the specified requirement.

4.4.2 Specific gravity of booster lead charge

4.4.2.1 Preformed pellets.-The specific gravity of the lead charge preformed pellet shall be determined using the following formula:

$$\text{Specific Gravity} = \frac{W}{12.8727DL^2}$$

Where:
W = weight of the pellets in grams
D = diameter of pellet in inches squared
L = length of pellet in inches

To assure the accuracy of the above formula, it shall be necessary that the diameter (D) and length (L) of the pellet be measured to the nearest one thousandth of an inch on an optical comparator. In addition, pellets containing chips or broken edges shall not be used for this determination.

4.4.2.2 Reconsolidated pellet in lead cup.-The specific gravity of the lead cup charge shall be determined as specified herein. All weights shall be measured to the nearest 0.001 gram, and all volumes calculated to the nearest 0.001 cubic centimeter. The resultant specific gravity calculation shall be rounded off to the nearest 0.01 by increasing the result if the thousandth place is greater than 5, increased if the hundredths place is an even number and the thousandths place is exactly 5, and unchanged if the hundredths place is odd and the thousandths place is exactly 5. This test shall be performed with water and ambient temperature at 72 degrees Fahrenheit, plus or minus 2 degrees Fahrenheit.

The procedure for weighing articles while submerged in water shall be as follows:

a. Fill a glass beaker with distilled water to approximately one inch of the top.

b. Attach a wire loop from the hanger of either balance pan of an analytical balance (without dampers). The wire loop shall be formed so that it will hold and suspend the lead cup under approximately one inch of water in the beaker mentioned above, when the balance is balanced, a tare weight shall be attached to the other balance pan to balance the wire loop when

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the wire loop, without the lead cup, is submerged to the depth required to submerge the lead cup. For analytical balances that do not utilize pan, or are of special design, a comparable procedure shall be established. The height of the water shall be maintained throughout the test.

c. Place the lead cup in the loop, open end up, and check to assure that all bubbles have been removed from the exterior surfaces when loaded and interior surfaces when empty. Balance and record this weight to the nearest 0.001 gram.

To determine the specific gravity of the Comp A-5 lead charge proceed as follows:

1) Weigh the loaded lead cup in air, record this weight as A.

2) Weigh the loaded lead cup under water, record this weight as B.

3) Subtract B from A and multiply by 0.9977. Record the result as the volume of the loaded lead cup, Vl.

4) Remove the Comp A-5 charge from the lead cup by soaking and washing with chloroform and acetone. After all visible traces of the Comp A-5 have been removed, dry the lead cup, weigh in air, and record this weight as C.

5) Weigh the empty lead cup under water and record this weight as D.

6) Subtract D from C and multiply by 0.0077. Record the result as the volume of the empty lead cup Ve.

7) Subtract Ve from Vl and record the result as the volume of Comp A-5 (Pv).

8) Subtract C from A and record the result as the weight of Comp A-5 (Pw).

9) Divide the weight of the Comp A-5 (Pw) by the volume of the Comp A-5 (Pv) and record the result as the specific gravity of the lead cup charge.

4.4.3 Determination of binder-lubricant content of booster lead charge.-An accurately weighed, moisture-free portion of 5 grams of sample shall be transferred to a tared, medium porosity, sintered glass filtering crucible. The sample shall be extracted with four 10-milliliter (ml.) portions of acetone; each portion

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shall remain incontact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible shall be washed down with each 10 ml. portion of chloroform and acetone to insure complete removal of the Comp A-5. The residue shall be aspirated until the odor of chloroform and acetone is no longer detectable. The crucible and its contents shall be placed in an oven maintained at 105 degrees plus or minus 5 degrees Centigrade (c) for 1/2 hour and then cooled to room temperature in a desiccator and then weighed. The residue shall then be reextracted with 10-ml. portions of chloroform and acetone, repeating the suction, drying, cooling and weighing operations after each additional portion until two successive weighings do not differ from each other by more than 0.002 grams. Calculate the total percentage of binder-lubricant as follows:

$$\text{Total percent of binder-lubricant} = \frac{100A}{B}$$

Where A - Weight of residue in crucible
B = Weight of specimen on a moisture-free basis

NOTE: The sample, to be moisture-free, should be placed in an ambient temperature of 100 plus or minus 5 degrees Centigrade for one hour and the weight recorded as B.

4.4.4 Non-arming.-The fuze metal parts assembly, without shield shall be mounted in a centrifuge with the longitudinal axis parallel to the rotating arm and the flanged end of the fuze pointing away from the center of rotation. The centrifuge shall be rotated at the required non-arming speed, and then stopped and the assembly removed and inspected for evidence that the fuze assembly is secured in the unarmed position. Such evidence shall be achieved by 100 percent examination by means of a 3 to 5 power magnification to determine non-arming by the following criteria.

a. Visual examination in unarmed position. The assembly shall be considered unarmed when the rotor pin is still retained by the leaf restraining point as indicated by View A and Note 7 of dwg. 9323857. (Attempt to release rotor by pushing with a wooden probe).

b. Rotor spring properly engaged in rotor shaft slot.

c. The pins in leaves Number 2 and 3 are in the forward flat positions of leaves 1 and 2 respectively. Particular

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observation shall be made to determine that leaf Number 1 is not inverted so that the forward flat surface is above the pin on Leaf Number 3.

d. Leaf Number 3 engagement under the anti-reset spring shall be approximately 2/64 inch. If engagement is approximately 1/64 inch or less, the assembly shall be rejected and removed from the lot.

4.4.5 Functioning.-The fuzes shall be assembled to the appropriate cartridge and fired at ambient temperatures at 5-inch homogeneous armor plate placed in a vertical position 400 feet from the muzzle of the gun. In the case of the M344A1 only, if 5-inch plate is not available, 4-inch, 3-inch, or 2-inch plate may be substituted. In the case of the M456A1E1 only, if 5-inch plate is not available, 6-inch plate may be substituted. Observation shall be made for fuze functioning and penetration by jet action, (see 6.7).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C - The fuzes shall be packaged in accordance with dwg. 9248976.

5.2 Packing

5.2.1 Level C - The fuzes shall be packed in accordance with dwg. 8837807.

5.3 Marking.-Marking shall be in accordance with dwg. 9248976.

6. NOTES

6.1 Intended use.-The fuze covered by this specification is intended for use in Cartridge, 105MM, HEAT-T, M456A1E1 and Cartridge, 106MM, HEAT, M344A1.

6.2 Ordering data (see 6.1 of MIL-A-48078 (PA).

6.3 Submission of design approval - see 6.2 of MIL-A-48078.-Submit equipment designs, as required to Commander, ARRADCOM, Hqts, Dover, N.J. 07801, ATTN: DRDAR-QA-A-F.

6.4 Non-arming.-The assembly shall be considered unarmed when the rotor pin is still retained by the leaf restraining point.

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6.5 Centrifugal acceleration.-The centrifugal acceleration expressed in gravities (g's) is calculated as follows:

$$a = .0000284\ (N)^2 r$$

Where:
a = acceleration expressed in g's
r = distance in inches from the axis of the centrifuge to the center of the release mechanism (see Note)
n = revolutions per minute of the centrifuge

NOTE: The distance between the center of the rotor shaft and the center of the release mechanism is 0.218 inch.

6.6 Submission of test data and ammunition data cards.-In addition to the normal distribution of records, copies of all ballistic data and ammunition data cards will be forwarded to the following: Commander, ARRADCOM, Hqts, Dover, N.J. 07801, ATTN: DRDAR-QA-A-F and DRDAR-LCR-E.

6.7 Penetration.-A "complete penetration" is defined as a hole produced by jet action extending completely through the plate. The lodging of a copper slug in the completed hole is not considered cause for rejection. Penetration produced by jet action is characterized by a relatively smooth tapered hole having a larger entrance than exit. A "partial penetration" occurs when a hole produced by jet action does not extend completely through the plate. The term "dud" shall be applied to rounds which merely break-up on the armor target, those which deflagrate (function low order), or those which penetrate by Kinetic energy. Rounds which deflagrate are characterized by a disk shaped cavity on the front face of the armor which may or may not be accompanied by spalling of the rear surface. Another possible indication of a dud is a large hole approximately the diameter of the projectile, which has been punched through the plate by Kinetic energy of the projectile with no evidence of jet action. Test reports shall distinguish between duds which merely break-up with no evidence of explosion or burning and duds which deflagrate or penetrate by Kinetic energy.

Custodian:
Army-AR

Preparing Activity:
Army-AR

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